# 1.6 New functions from old functions

Translations :

Vertical and horizontal shifts :

Suppose $c > 0$

$y = f(x) + c$, shift the graph of $y = f(x)$ a distance $c$ units upward

$y = f(x) - c$ — " — $c$ units downward

$y = f(x - c)$ — " — $c$ units to the right

$y = f(x + c)$ — " — $c$ units to the left

Streching and reflections :

Vertical and horizontal stretching and reflecting.

Suppose $c > 1$. To obtain the graph of

• $y = c \cdot f(x)$, strech the graph of $y = f(x)$ vertically by a factor $c$

- $y = \frac{1}{c} \cdot f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of $c$
- $y = f(cx)$, shrink the graph $y = f(x)$ horizontally by a factor of $c$
- $y = f\left(\frac{x}{c}\right)$, strech —— " —— horizontally by a factor of $c$
- $y = -f(x)$, reflect —— " —— about the $x$-axis
- $y = f(-x)$, —— " —— about the $y$-axis

## Combinations of functions

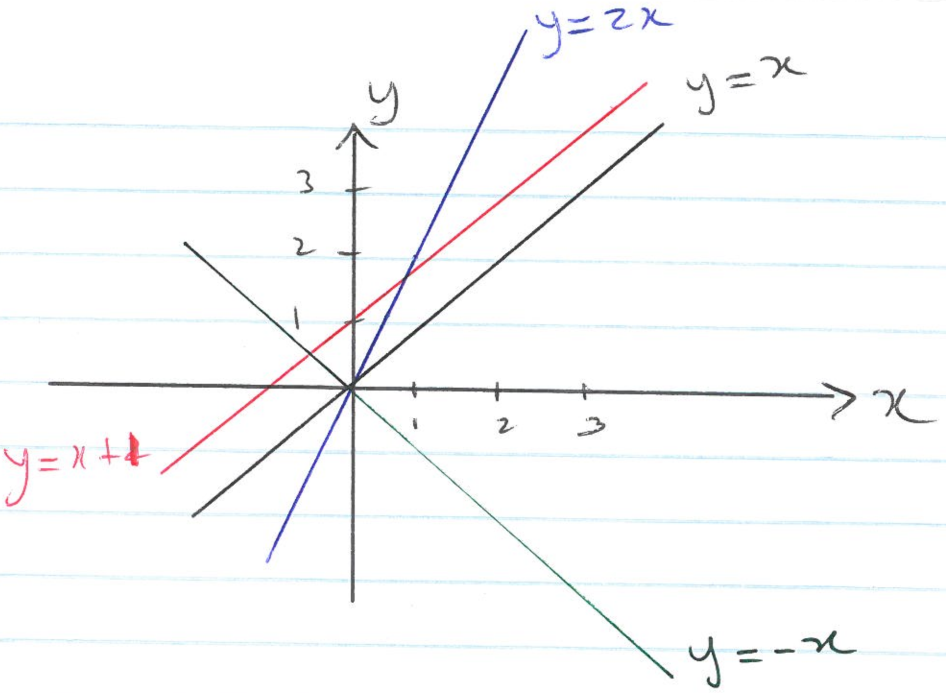
Two functions $f$ and $g$ can be combined to form new functions

$$(f+g)(x) = f(x) + g(x)$$

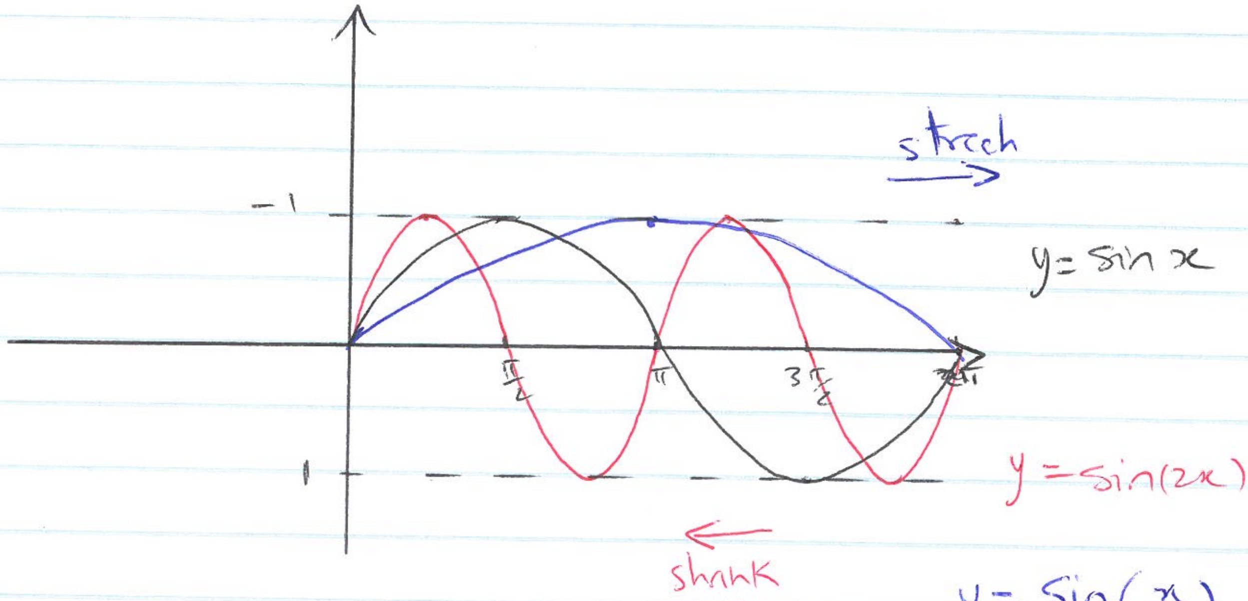
$$(f-g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

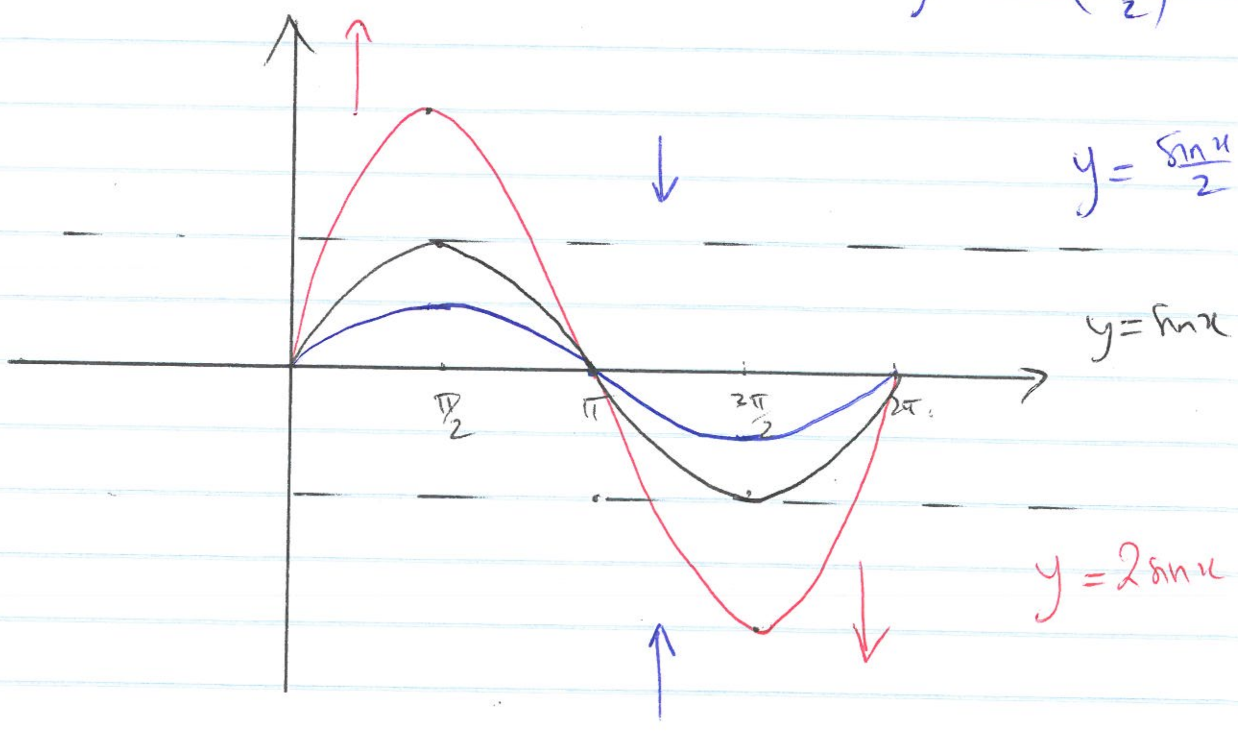
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$



y = 2x
y = x
y
3
2
1
x
1
2
3
y = x + 1
y = -x

strech
-1
y = sin x
π/2
π
3π/2
2π
1
y = sin(2x)
shrink
y = sin(x/2)

y = sin x / 2
y = sin x
π/2
π
3π/2
2π
y = 2 sin x

The domain of $f+g$, $f-g$ and $f.g$ is $D_f \cap D_g$, whereas the domain of $\frac{f}{g}$ is $D_f \cap D_g \setminus \{x / g(x) = 0\}$

$$(f \circ g)(x) = f(g(x))$$

$$D_{f \circ g} = \{x \in \mathbb{R} / g(x) \in D_f\}$$

In general $f \circ g \neq g \circ f$.

Example 0 :

(a) $h(x) = \frac{1}{\sin x}$, $D_h = \{x \in \mathbb{R} / \sin x \neq 0\}$

$= \{x \in \mathbb{R} / x \neq k\pi, k \in \mathbb{Z}\}$

$h = f \circ g$ with $f(x) = \frac{1}{x}$ and $g(x) = \sin x$

(b) $h(x) = \sqrt{x^2 - 1} = (f \circ g)(x)$ with $f(x) = \sqrt{x}$

and $g(x) = x^2 - 1$

$D_h = \{x \in \mathbb{R} / g(x) \geq 0\} = (-\infty, -1] \cup [1, +\infty)$

Example 1: Write down $f \circ g$ and $g \circ f$ given

(i) $f(x) = \tan x$ , $g(x) = x^2$

(ii) $f(x) = 2^x$ , $g(x) = \log_2 x$

(iii) $f(x) = x^3 + \frac{1}{2x}$ , $g(x) = \cos x$

: (i) $(f \circ g)(x) = \tan(x^2)$

$(g \circ f)(x) = \tan^2 x$

(ii) $(f \circ g)(x) = 2^{\log_2 x} = x$

$(g \circ f)(x) = \log_2(2^x) = x$

(iii) $(f \circ g)(x) = \cos^3 x + \frac{1}{2\cos x}$

$(g \circ f)(x) = \cos\left(x^3 + \frac{1}{2x}\right)$

**Example 2** Given $F(x)$, find $f$ and $g$ such that $F = f \circ g$

(i) $F(x) = \dfrac{1}{x^2+2}$

(ii) $F(x) = \cos^3(\sqrt{x})$

**Example 3**: Write down $f$, $g$ and $h$ such that $F = f \circ g \circ h$ if

(i) $F(x) = \cos^3(\sqrt{x})$ ; (ii) $F(x) = \dfrac{1}{2^{3-x}}$

**Eg 2**: (i) $f(x) = \dfrac{1}{x+2}$, $g(x) = x^2$

or $f(x) = \dfrac{1}{x}$, $g(x) = x^2+2$

(ii) $f(x) = x^3$, $g(x) = \cos(\sqrt{x})$

or $f(x) = \cos^3(x)$, $g(x) = \sqrt{x}$

**Eg 3**: (i) $f(x) = x^3$, $g(x) = \cos x$, $h(x) = \sqrt{x}$

(ii) $f(x) = \dfrac{1}{x}$, $g(x) = 2^x$, $h(x) = 3-x$